

Corrected structured proof II

Uniform gap and local moment $\implies$ local-norm convergence and the main theorem

Fresh reconstruction of `part_i/main.tex`

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Status. CONDITIONAL. Given the explicitly listed inputs in Section 1, the full cutoff net is Cauchy in the norm of every local dual, with a super-polynomial rate. The resulting state is locally normal, translation invariant, parity invariant for even interaction, and a spectral ground state.

The derivation below is independent of Euclidean uniqueness. The current Part I document correctly states its two-phase exclusion only conditionally on the load-bearing phase-decomposition hypothesis B4; the final section records that scope.

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1 Exact inputs

For a spatial cutoff $g \in \mathcal{G}$, let

$$K(g) = H_0 + V(g), \quad E(g) = \inf \text{spec } K(g), \quad H(g) = K(g) - E(g), \quad P_g = |\Omega_g\rangle\langle\Omega_g|,$$

and let $\omega_g(A) = \langle\Omega_g, A\Omega_g\rangle$ and $\beta_t^g = \text{Ad } e^{itH(g)}$. Let $\mathfrak{A}(O)$ be the time-zero local von Neumann algebra of a bounded interval O and let O_r be its open r -neighbourhood.

Imported assumption 1.1 (FKN, stability, and free hypercontractivity (F)). For every affine cutoff path $g_s = (1-s)g + sg'$ and every $T > 0$, the interacting FKN formula (3.1) below holds. Its finite-slab interaction random variables satisfy $\mathcal{U}_v \in L^p$ for every finite p and $e^{-a\mathcal{U}_g} \in L^1$ for every finite $a > 0$. The free time-zero Markov semigroup satisfies Nelson hypercontractivity with $(q-1)/(p-1) = e^{2m_0T}$. These are the precise standard inputs used to prove path regularity; they are not consequences of (G) or (M).

Imported assumption 1.2 (uniform physical gap (G)). There is $\gamma > 0$, independent of g , such that

$$\text{spec } H(g) \subseteq \{0\} \cup [\gamma, \infty), \quad \dim \ker H(g) = 1. \quad (\text{G})$$

Imported assumption 1.3 (uniform local interaction moment (M)). There is $M < \infty$ such that

$$\|V(h)\Omega_g\| \leq M \quad (\text{M})$$

for every g and every real $h \in C_c^\infty(\mathbb{R})$ satisfying $\|h\|_\infty \leq 1$ and whose support is contained in an interval of length 2.

Imported assumption 1.4 (local domains and affiliation). For every compactly supported real h , $\Omega_g \in D(V(h))$, and $V(h)$ is self-adjoint and affiliated with $\mathfrak{A}(U)$ whenever U contains $\text{supp } h$. Along every affine path there is a common core $\mathcal{C}$ for the K_s such that $K_s\xi = K_{s_0}\xi + (s - s_0)V(g' - g)\xi$ for $\xi \in \mathcal{C}$. The correct model-specific smoothing source is Glimm–Jaffe Expositions, Theorem 7.4. Theorem 7.3 alone assumes that the multiplication potential is bounded below and therefore does not, by itself, justify this application.

Imported assumption 1.5 (finite propagation and patching). For every cutoff,

$$\beta_t^g(\mathfrak{A}(O)) \subseteq \mathfrak{A}(O_{|t|}). \quad (\text{FP})$$

There is a cutoff-independent quasi-local dynamics α such that

$$g = 1 \text{ on } O_{|t|} \implies \alpha_t(A) = \beta_t^g(A) \quad (A \in \mathfrak{A}(O)). \quad (\text{P})$$

Imported assumption 1.6 (covariance). If $(\tau_a g)(x) = g(x - a)$, then

$$\omega_{\tau_a g} = \omega_g \circ \tau_{-a}. \quad (\text{T})$$

For even $\mathcal{P}$, each ω_g is invariant under $\theta = \text{Ad } \Gamma(-1)$.

The first corrected proof document derives Assumptions 1.2 and 1.3 from GJS in the weak-coupling region. Assumptions 1.1 and 1.4–1.6 are standard but logically independent cutoff-construction inputs.

2 Statement to be proved

For a cutoff g and interval O , write

$$d(g, O) = \text{dist}(O, \mathbb{R} \setminus \{g = 1\}^\circ).$$

Theorem 2.1 (local-norm convergence and spectral ground state). *Under Assumptions 1.1–1.6, there is a state ω_∞ on the quasi-local algebra $\mathfrak{A}$ such that, for every bounded interval O ,*

$$\lim_{g \in \mathcal{G}} \|(\omega_g - \omega_\infty)|_{\mathfrak{A}(O)}\| = 0. \quad (2.1)$$

If $g = 1$ on O_d , then

$$\|(\omega_g - \omega_\infty)|_{\mathfrak{A}(O)}\| \leq \Psi_\gamma(d), \quad (2.2)$$

where $\Psi_\gamma(d) = O_N((1 + d)^{-N})$ for every N . The limit is locally normal, is a spectral ground state for $(\mathfrak{A}, \alpha)$, is translation invariant, and, for even $\mathcal{P}$, is θ -invariant.

3 Structured proof

⟨1⟩ 1 Prove differentiability of the ground-state projection along an affine cutoff path.

Let g, g' equal 1 on a common nonempty interval, put

$$g_s = (1-s)g + sg', \quad v = g' - g, \quad K_s = K(g_s), \quad H_s = K_s - E_s, P_s = |\Omega_s\rangle\langle\Omega_s|, \quad Q_s = 1 - P_s.$$

⟨2⟩ 1 The family $s \mapsto e^{-TK_s}$ is norm-continuous on $[0, 1]$ and norm- C^1 on $(0, 1)$ for each $T > 0$.

Because: On free Euclidean path space define

$$\mathcal{U}_s = \lambda \int_0^T \int g_s(x) : \mathcal{P}(\Phi(t, x)) : dx dt, \quad w_s = e^{-\mathcal{U}_s}, \quad \mathcal{U}_v = \mathcal{U}_1 - \mathcal{U}_0.$$

For bounded time-zero f, h , the interacting FKN formula is

$$\langle f, e^{-TK_s} h \rangle = \int \overline{J_T f} J_0 h w_s d\mu_0. \quad (3.1)$$

The correct references are Simon Corollary V.14/Theorem V.15 or GRS Theorem II.16, not Simon Theorem III.12.

Set $a = e^{-m_0 T}$ and

$$q_T = \frac{2}{1-a}, \quad r_T = \frac{2}{1+a}, \quad p_T = 1+a, \quad p'_T = 1+a^{-1}.$$

Then $1/r_T + 1/q_T = 1$, $r_T p_T = 2$, and $(p'_T - 1)/(p_T - 1) = e^{2m_0 T}$. Nelson hypercontractivity—Simon Theorem I.17, or its time-region version Theorem III.17—therefore gives

$$\|P_T F\|_{p'_T} \leq \|F\|_{p_T}.$$

The Markov property and Hölder yield

$$\begin{aligned} \|(J_T f)(J_0 h)\|_{r_T}^{r_T} &= \int |h|^{r_T} P_T(|f|^{r_T}) d\nu_0 \\ &\leq \| |h|^{r_T} \|_{p_T} \|P_T(|f|^{r_T})\|_{p'_T} \\ &\leq \|h\|_2^{r_T} \|f\|_2^{r_T}. \end{aligned}$$

Thus

$$\|(J_T f)(J_0 h)\|_{r_T} \leq \|f\|_2 \|h\|_2. \quad (3.2)$$

Finite-volume stability gives $\mathcal{U}_v \in L^p(\mu_0)$ for every finite p and $e^{-a\mathcal{U}_i} \in L^1(\mu_0)$ for every $a < \infty$, $i = 0, 1$ (Simon Theorem V.7). Since $\mathcal{U}_s = (1-s)\mathcal{U}_0 + s\mathcal{U}_1$, Hölder gives

$$\|w_s\|_a^a \leq \left(\int e^{-a\mathcal{U}_0} d\mu_0 \right)^{1-s} \left(\int e^{-a\mathcal{U}_1} d\mu_0 \right)^s. \quad (3.3)$$

Convergence in probability together with (3.3) at a larger exponent gives $w_{s_n} \rightarrow w_s$ in L^a by Vitali. Moreover

$$\frac{w_{s+h} - w_s}{h} = -\mathcal{U}_v \int_0^1 w_{s+\vartheta h} d\vartheta. \quad (3.4)$$

Choose finite p, a with $q_T^{-1} = p^{-1} + a^{-1}$. Hölder and the just-proved L^a continuity show that (3.4) converges in L^{q_T} to $-\mathcal{U}_v w_s$ for $0 < s < 1$. Finally (3.1), (3.2), and duality give

$$\|e^{-TK_s} - e^{-TK_r}\| \leq \|w_s - w_r\|_{q_T},$$

and the same estimate for difference quotients. This proves the asserted norm regularity.

$\langle 2 \rangle 2$ $s \mapsto P_s$ is norm-continuous on $[0, 1]$ and C^1 on $(0, 1)$.

Because: $L_s = e^{-TK_s}$ has the simple largest eigenvalue $\lambda_s = e^{-TE_s} = \|L_s\|$. By (G),

$$\text{spec } L_s \setminus \{\lambda_s\} \subset [0, e^{-T\gamma}\lambda_s].$$

The relative spectral separation $1 - e^{-T\gamma}$ is uniform. A Riesz contour around λ_s , together with the norm regularity from $\langle 2 \rangle 1$, gives exactly the stated regularity of P_s .

$\langle 2 \rangle 3$ Locally on $(0, 1)$ choose normalized C^1 vectors in the parallel gauge $\langle \Omega_s, \dot{\Omega}_s \rangle = 0$. Then

$$\dot{E}_s = \langle \Omega_s, V(v)\Omega_s \rangle, \quad \dot{\Omega}_s = -H_s^{-1}Q_s V(v)\Omega_s. \quad (3.5)$$

Because: Let $\mathcal{C}$ be the common core supplied by cutoff construction. For $\xi \in \mathcal{C}$, $K_s \xi = K_{s_0} \xi + (s - s_0)V(v)\xi$. Differentiate the weak eigenvalue identity $\langle K_s \xi, \Omega_s \rangle = E_s \langle \xi, \Omega_s \rangle$ to obtain

$$\langle H_s \xi, \dot{\Omega}_s \rangle = \langle \xi, (\dot{E}_s - V(v))\Omega_s \rangle. \quad (3.6)$$

Assumption 1.4 makes the vector on the right well defined. Since $\mathcal{C}$ is a core for the self-adjoint H_s , (3.6) says $\dot{\Omega}_s \in D(H_s)$ and $H_s \dot{\Omega}_s = (\dot{E}_s - V(v))\Omega_s$. Pairing with Ω_s proves the first formula in (3.5). The parallel gauge puts $\dot{\Omega}_s$ in $Q_s \mathcal{H}$; (G) makes $H_s^{-1}Q_s$ bounded by γ^{-1} , proving the second formula.

$\langle 2 \rangle 4$ This proves $\langle 1 \rangle 1$. □

$\langle 1 \rangle 2$ Construct an exact inverse-energy filter with super-polynomial time tail.

$\langle 2 \rangle 1$ Choose even $\chi \in C^\infty(\mathbb{R}; [0, 1])$ with $\chi = 0$ on $[-\gamma/2, \gamma/2]$ and $\chi = 1$ outside $(-\gamma, \gamma)$, and set

$$\widehat{W}_\gamma(E) = \begin{cases} \chi(E)/E, & E \neq 0, \\ 0, & E = 0. \end{cases} \quad (3.7)$$

$\langle 2 \rangle 2$ Its inverse transform $W_\gamma(t) = (2\pi)^{-1} \int e^{-itE} \widehat{W}_\gamma(E) dE$ is an L^1 function satisfying

$$\int (1 + |t|)^q |W_\gamma(t)| dt < \infty \quad (q \geq 0). \quad (3.8)$$

Because: Oddness gives

$$W_\gamma(t) = -\frac{i}{\pi} \int_0^\infty \frac{\sin(tE)\chi(E)}{E} dE.$$

For $0 < |t| \leq 1$, the compact transition part is absolutely bounded and the tail is uniformly bounded after $u = |t|E$ by Dirichlet's test. For $|t| \geq 1$, every derivative $\widehat{W}_\gamma^{(k)}$ with $k \geq 1$ is integrable, so k integrations by parts give

$$|W_\gamma(t)| \leq \frac{\|\widehat{W}_\gamma^{(k)}\|_1}{2\pi|t|^k}.$$

Choose $k > q + 1$. This proves (3.8), including integrability near zero.

$\langle 2 \rangle 3$ If H satisfies (G), then

$$H^{-1}(1 - P_{\ker H}) = \int_{\mathbb{R}} W_\gamma(t) e^{itH} (1 - P_{\ker H}) dt. \quad (3.9)$$

Because: Fourier inversion gives $\widehat{W}_\gamma(E) = E^{-1}$ on $[\gamma, \infty)$ and zero at $E = 0$. The spectral theorem gives (3.9); it is a norm-convergent Bochner integral because $W_\gamma \in L^1$.

Define

$$T_\gamma(r) = \int_{|t|>r} |W_\gamma(t)| dt.$$

From (3.8), Markov's inequality gives $T_\gamma(r) \leq C_{q,\gamma}(1+r)^{-q}$ for every q .

⟨2⟩ 4 This proves ⟨1⟩2. □

⟨1⟩ 3 Prove the spectral-separation estimate used for clustering.

Suppose

$$F(t) = \int_{[\gamma, \infty)} e^{itE} d\rho(E), \quad G(t) = \int_{(-\infty, -\gamma]} e^{itE} d\sigma(E),$$

with $|\rho|(\mathbb{R}), |\sigma|(\mathbb{R}) \leq M_0$, and $F = G$ on $(-R, R)$, where $\gamma R \geq 1$.

⟨2⟩ 1 For $a, \varepsilon > 0$ define

$$f_{a,\varepsilon}(t) = \frac{1}{2\pi i} \frac{e^{-t^2/(2a)}}{t - i\varepsilon}.$$

Its transform is

$$\check{f}_{a,\varepsilon}(E) = \int_0^\infty \sqrt{\frac{a}{2\pi}} e^{-a(E-E')^2/2} e^{-\varepsilon E'} dE', \quad (3.10)$$

which lies in $[0, 1]$ and tends to the standard normal distribution function $\Phi(\sqrt{a}E)$ as $\varepsilon \downarrow 0$.

Because: The transform of $(2\pi i)^{-1}(t - i\varepsilon)^{-1}$ is $\mathbf{1}_{(0,\infty)}(E)e^{-\varepsilon E}$ by a one-pole contour integral. Multiplication by the Gaussian in t convolves this function with the normalized Gaussian displayed in (3.10).

⟨2⟩ 2 The part of $f_{a,\varepsilon}$ outside $(-R, R)$ satisfies

$$\int_{|t|\geq R} |f_{a,\varepsilon}(t)| dt \leq \frac{a}{\pi R^2} e^{-R^2/(2a)}. \quad (3.11)$$

Because: $|t - i\varepsilon| \geq |t|$ and, for $t \geq R$, $t^{-1} \leq t/R^2$. Integrating $te^{-t^2/(2a)}$ exactly proves (3.11).

⟨2⟩ 3 Using $F = G$ on $(-R, R)$ and Fubini gives

$$\left| \int \Phi(\sqrt{a}E) d\rho(E) - \int \Phi(\sqrt{a}E) d\sigma(E) \right| \leq \frac{2M_0a}{\pi R^2} e^{-R^2/(2a)}. \quad (3.12)$$

Because: Before $\varepsilon \downarrow 0$, the difference of the two time integrals is supported in $|t| \geq R$ and is at most $(|\rho|(\mathbb{R}) + |\sigma|(\mathbb{R}))$ times (3.11). Dominated convergence in energy proves (3.12).

⟨2⟩ 4 Therefore

$$|F(0)| \leq 2M_0 e^{-\gamma R/2}. \quad (3.13)$$

Because: Insert and subtract the two integrals in (3.12). On the supports, $1 - \Phi(\sqrt{a}E) \leq \frac{1}{2}e^{-a\gamma^2/2}$ and $\Phi(\sqrt{a}E) \leq \frac{1}{2}e^{-a\gamma^2/2}$, respectively. Hence

$$|F(0)| \leq M_0 e^{-a\gamma^2/2} + \frac{2M_0a}{\pi R^2} e^{-R^2/(2a)}.$$

Set $a = R/\gamma$. Both exponentials become $e^{-\gamma R/2}$ and $2a/(\pi R^2) = 2/(\pi\gamma R) \leq 2/\pi < 1$, which proves (3.13).

⟨2⟩ 5 This proves ⟨1⟩3. □

⟨1⟩ 4 Derive clustering with one unbounded local observable.

Let H, Ω satisfy (G), let $X = X^*$ be affiliated with every $\mathfrak{A}(U)$ containing a compact set B , assume $\Omega \in D(X)$, and let $A \in \mathfrak{A}(O)$. Put $R = \text{dist}(B, O) > 0$ and suppose $\gamma R \geq 1$.

⟨2⟩ 1 Define

$$\begin{aligned} F(t) &= \langle X\Omega, (e^{itH} - P_0)A\Omega \rangle, \\ G(t) &= \langle A^*\Omega, (e^{-itH} - P_0)X\Omega \rangle, \quad P_0 = |\Omega\rangle\langle\Omega|. \end{aligned}$$

Then F and G have spectral measures supported in $[\gamma, \infty)$ and $(-\infty, -\gamma]$, with variations at most $M_0 = \|X\Omega\| \|A\|$.

Because: Insert the spectral resolution of H on $(1 - P_0)\mathcal{H}$ and use Cauchy–Schwarz for the variation of the resulting complex measures.

⟨2⟩ 2 $F(t) = G(t)$ for $|t| < R$.

Because: By finite propagation, $\beta_t(A) \in \mathfrak{A}(O_{|t|})$. Choose a bounded open $U \supset B$ disjoint from $O_{|t|}$. Affiliation implies that $\beta_t(A) \in \mathfrak{A}(U)'$ preserves $D(X)$ and commutes with X there. Since $e^{-itH}\Omega = \Omega$,

$$\begin{aligned} \langle X\Omega, e^{itH}A\Omega \rangle &= \langle \Omega, X\beta_t(A)\Omega \rangle = \langle \Omega, \beta_t(A)X\Omega \rangle \\ &= \langle A^*\Omega, e^{-itH}X\Omega \rangle. \end{aligned}$$

The P_0 terms agree because $\langle \Omega, X\Omega \rangle$ is real.

⟨2⟩ 3 Apply (3.13) to obtain

$$|\langle X\Omega, A\Omega \rangle - \langle X\Omega, \Omega \rangle \langle \Omega, A\Omega \rangle| \leq 2\|X\Omega\| \|A\| e^{-\gamma R/2}. \quad (3.14)$$

⟨2⟩ 4 This proves ⟨1⟩4. □

⟨1⟩ 5 Differentiate the state and localize the perturbation.

⟨2⟩ 1 For self-adjoint $A \in B(\mathcal{H})$ and $0 < s < 1$,

$$\frac{d}{ds}\omega_{g_s}(A) = -2 \operatorname{Re} \int_{\mathbb{R}} W_\gamma(t) [\langle V(v)\Omega_s, \beta_t^{g_s}(A)\Omega_s \rangle - \langle V(v)\Omega_s, \Omega_s \rangle \omega_{g_s}(A)] dt. \quad (3.15)$$

Because: Differentiate the vector state and insert (3.5):

$$\frac{d}{ds}\omega_{g_s}(A) = -2 \operatorname{Re} \langle V(v)\Omega_s, H_s^{-1}Q_s A\Omega_s \rangle.$$

Insert the exact filter (3.9). Since $e^{itH_s}Q_s A\Omega_s = \beta_t^{g_s}(A)\Omega_s - \omega_{g_s}(A)\Omega_s$, this is (3.15). The integral is absolutely convergent because $W_\gamma \in L^1$ and $V(v)\Omega_s$ exists; v has compact support.

⟨2⟩ 2 Choose a smooth partition of unity $\sum_j \eta_j = 1$ with $0 \leq \eta_j \leq 1$ and $\operatorname{supp} \eta_j \subset (j-1, j+1)$. Put $h_j = \eta_j v$, $V_j = V(h_j)$. Then

$$V(v)\xi = \sum_j V_j \xi, \quad \|V_j \Omega_s\| \leq M. \quad (3.16)$$

Because: Only finitely many h_j are nonzero. Also $|v| \leq 1$, so $\|h_j\|_\infty \leq 1$, and each support has length at most 2. Apply (M) to the cutoff g_s .

⟨2⟩ 3 This proves ⟨1⟩5. □

⟨1⟩ 6 Prove the one-path local-perturbations estimate.

Assume $g = g' = 1$ on O_d , where

$$d \geq d_0 := 4 + 2/\gamma.$$

For a nonzero h_j set $B_j = \text{supp } h_j$ and $r_j = \text{dist}(B_j, O)$.

⟨2⟩ 1 $r_j \geq d$ for every nonzero h_j .

Because: On O_d , $v = g' - g = 0$. Therefore the support of $h_j = \eta_j v$ is contained in $\mathbb{R} \setminus O_d$.

⟨2⟩ 2 For self-adjoint $A \in \mathfrak{A}(O)$, $\|A\| \leq 1$, define

$$C_{j,s}(t) = \langle V_j \Omega_s, \beta_t^{g_s}(A) \Omega_s \rangle - \langle V_j \Omega_s, \Omega_s \rangle \omega_{g_s}(A).$$

Then

$$|C_{j,s}(t)| \leq \begin{cases} 2M e^{-\gamma r_j/4}, & |t| \leq r_j/2, \\ 2M, & t \in \mathbb{R}. \end{cases} \quad (3.17)$$

Because: In the first case, finite propagation places $\beta_t^{g_s}(A)$ in $\mathfrak{A}(O_{|t|})$ and $\text{dist}(B_j, O_{|t|}) \geq r_j/2$. Moreover $\gamma r_j/2 \geq 1$. Apply (3.14) with $X = V_j$, then (3.16). For arbitrary t , bound each of the two terms defining $C_{j,s}(t)$ by M .

⟨2⟩ 3 Equations (3.15)–(3.17) imply

$$\left| \frac{d}{ds} \omega_{g_s}(A) \right| \leq 4M \sum_j \left(\|W_\gamma\|_1 e^{-\gamma r_j/4} + T_\gamma(r_j/2) \right). \quad (3.18)$$

Because: Split the t -integral for each j at $|t| = r_j/2$. The outside factor 2 in (3.15), the factor $2M$ in either line of (3.17), and the two integral regions give exactly the coefficient $4M$ in (3.18).

⟨2⟩ 4 At most eight indices j have $r_j \in [n, n+1)$ for any integer $n \geq \lfloor d \rfloor$.

Because: Write $O = (a, b)$. Since $d \geq 4$ and $B_j \subset [j-1, j+1]$, a nonempty B_j lies wholly on one side of O_d . On the right, compactness gives $x_j \in B_j$ with $x_j - b = r_j$. Since $|x_j - j| \leq 1$ and $r_j \in [n, n+1)$,

$$j \in [b + n - 1, b + n + 2],$$

which contains at most four integers. The left side gives four more.

⟨2⟩ 5 Integrating (3.18) along the path gives

$$\|(\omega_g - \omega_{g'})|_{\mathfrak{A}(O)}\| \leq 32M \sum_{n \geq \lfloor d \rfloor} \left(\|W_\gamma\|_1 e^{-\gamma n/4} + T_\gamma(n/2) \right). \quad (3.19)$$

Because: Both functions of r_j in (3.18) decrease with r_j . Apply the shell count from ⟨2⟩4, producing $4M \cdot 8 = 32M$. Integrate first on $[\varepsilon, 1 - \varepsilon]$, where differentiability holds, and then let $\varepsilon \downarrow 0$ using the endpoint norm continuity of P_s proved in ⟨1⟩1. Taking the supremum over self-adjoint contractions gives the norm of the hermitian functional $\omega_g - \omega_{g'}$.

⟨2⟩ 6 This proves ⟨1⟩6. □

⟨1⟩ 7 Define the rate and prove that it is super-polynomial.

For $d \geq d_0$ set

$$\Psi_\gamma(d) = 32M \sum_{n \geq \lfloor d \rfloor} \left(\|W_\gamma\|_1 e^{-\gamma n/4} + T_\gamma(n/2) \right), \quad (3.20)$$

and for $d < d_0$ enlarge it to at least 2.

⟨2⟩ 1 For every $N \geq 1$, there is $C_{N,\gamma,M}$ such that

$$\Psi_\gamma(d) \leq C_{N,\gamma,M}(1+d)^{-N}. \quad (3.21)$$

Because: The first series has an exponential tail. For the second choose $q = N + 2$ in $T_\gamma(r) \leq C_{q,\gamma}(1+r)^{-q}$; then $\sum_{n \geq [d]} (1+n)^{-N-2} = O((1+d)^{-N-1})$.

⟨2⟩ 2 This proves ⟨1⟩7. □

⟨1⟩ 8 Prove the full cutoff net is locally norm Cauchy and construct ω_∞ .

⟨2⟩ 1 Fix O and $\varepsilon > 0$. Choose $d \geq d_0$ with $\Psi_\gamma(d) < \varepsilon$, and choose $g_0 \in \mathcal{G}$ with $g_0 = 1$ on O_d .

⟨2⟩ 2 If $g, g' \succcurlyeq g_0$, then

$$\|(\omega_g - \omega_{g'})|_{\mathfrak{A}(O)}\| < \varepsilon. \quad (3.22)$$

Because: The directed order is inclusion of plateau interiors, so both cutoffs equal 1 on O_d . Apply (3.19).

⟨2⟩ 3 The limit functionals on local algebras are compatible and extend uniquely to a state ω_∞ on the norm-dense quasi-local union.

Because: Each $\mathfrak{A}(O)^*$ is complete, so (3.22) gives a limit on $\mathfrak{A}(O)$. If $O_1 \subset O_2$, uniqueness of the norm limit shows that the O_2 limit restricts to the O_1 limit. Positivity and normalization pass to the limit. A compatible bounded functional of norm one on the algebraic union extends uniquely by norm continuity to $\mathfrak{A}$.

⟨2⟩ 4 If one fixed g equals 1 on O_d , then (2.2) holds.

Because: The subset of cutoffs g' equal to 1 on O_d is cofinal. Apply (3.19) to (g, g') and let g' tend through this subset to the local norm limit.

⟨2⟩ 5 This proves local-norm convergence and its rate. □

⟨1⟩ 9 Prove local normality.

Fix O and let $0 \leq A_i \uparrow A$ in $\mathfrak{A}(O)$. For any cutoff g ,

$$0 \leq \omega_\infty(A) - \omega_\infty(A_i) \leq 2\delta\|A\| + \omega_g(A - A_i) \quad (3.23)$$

whenever $\|(\omega_g - \omega_\infty)|_{\mathfrak{A}(O)}\| < \delta$.

Because: Add and subtract $\omega_g(A)$ and $\omega_g(A_i)$, use $\|A_i\| \leq \|A\|$, and bound each state difference by the local dual norm. The cutoff vector state is normal on $\mathfrak{A}(O)$, so its last term in (3.23) tends to zero with i . First let i increase, then let $\delta \downarrow 0$. Thus ω_∞ is normal on $\mathfrak{A}(O)$.

⟨1⟩ 10 Prove the spectral ground-state condition.

Choose cutoffs g_n with $g_n = 1$ on $(-n, n)$. For local $A, B \in \mathfrak{A}(O)$ define the finite complex measure

$$\rho_n(S) = \langle A^* \Omega_{g_n}, \mathbf{1}_S(H(g_n)) B \Omega_{g_n} \rangle, \quad G_n(t) = \int_{[0, \infty)} e^{itE} d\rho_n(E). \quad (3.24)$$

⟨2⟩ 1 For $|t| < d(g_n, O)$,

$$\omega_{g_n}(A \alpha_t(B)) = G_n(t). \quad (3.25)$$

Because: Patching gives $\alpha_t(B) = \beta_t^{g_n}(B)$, and $e^{-itH(g_n)}\Omega_{g_n} = \Omega_{g_n}$. Moving the unitary to the middle gives (3.25).

$\langle 2 \rangle 2$ If $f \in L^1(\mathbb{R})$ and $\check{f}(E) = \int f(t)e^{itE}dt$ vanishes on $[0, \infty)$, then

$$\int_{\mathbb{R}} f(t)G_n(t)dt = 0. \quad (3.26)$$

Because: Fubini is justified by $|\rho_n|(\mathbb{R}) \leq \|A\|\|B\|$, and the spectral support in (3.24) is nonnegative.

$\langle 2 \rangle 3$ The limit function $G(t) = \omega_\infty(A\alpha_t(B))$ is continuous and satisfies $\int fG = 0$.

Because: For each T , eventually $d(g_n, O) > T$, and $A\alpha_t(B) \in \mathfrak{A}(O_T)$ for $|t| \leq T$. Hence

$$\sup_{|t| \leq T} |G(t) - G_n(t)| \leq \|A\|\|B\| \|(\omega_\infty - \omega_{g_n})|_{\mathfrak{A}(O_T)}\| \longrightarrow 0. \quad (3.27)$$

Thus G is a locally uniform limit of continuous functions. Let $M_0 = \|A\|\|B\|$ and $\epsilon_T = \int_{|t| > T} |f(t)|dt$. Using (3.26),

$$\left| \int fG \right| \leq \|f\|_1 M_0 \|(\omega_\infty - \omega_{g_n})|_{\mathfrak{A}(O_T)}\| + 2M_0 \epsilon_T.$$

First let $n \rightarrow \infty$, then $T \rightarrow \infty$.

$\langle 2 \rangle 4$ $\omega_\infty \circ \alpha_t = \omega_\infty$.

Because: For local A and fixed t , eventually patching gives $\omega_{g_n}(\alpha_t(A)) = \omega_{g_n}(A)$. Both observables lie in a fixed local algebra, so local norm convergence passes the identity to the limit. Density extends it to $\mathfrak{A}$.

$\langle 2 \rangle 5$ Steps $\langle 2 \rangle 3$ – $\langle 2 \rangle 4$ are exactly the spectral ground-state condition. □

$\langle 1 \rangle 11$ Prove spatial and parity invariance.

$\langle 2 \rangle 1$ $\omega_\infty \circ \tau_a = \omega_\infty$ for every $a \in \mathbb{R}$.

Because: $g \mapsto \tau_a g$ is an order isomorphism of the cutoff net and hence is cofinal. For local A , covariance (T) and local norm convergence give

$$\omega_\infty(A) = \lim_g \omega_{\tau_a g}(A) = \lim_g \omega_g(\tau_{-a}A) = \omega_\infty(\tau_{-a}A).$$

Extend by norm density.

$\langle 2 \rangle 2$ For even $\mathscr{P}$, $\omega_\infty \circ \theta = \omega_\infty$.

Because: Every cutoff state has this invariance by Assumption 1.6; pass to the local norm limit, then use density.

$\langle 2 \rangle 3$ Together with $\langle 1 \rangle 8$ – $\langle 1 \rangle 10$, this proves Theorem 2.1. □

4 The averaged Glimm–Jaffe vacuum

The manuscript’s canonical-vacuum corollary does follow. Let $g_n(x) = g(x/n)$ with $g = 1$ on $[-3, 3]$, and let $h \geq 0$, $\text{supp } h \subset [-1, 1]$, $\int h = 1$. The averaged state of Glimm–Jaffe III has the form

$$\omega_n(A) = \int_{-1}^1 h(u)\omega_{\tau_{-enu}g_n}(A)du, \quad \epsilon \in \{\pm 1\}.$$

For $O \subset [-R, R]$, every shifted cutoff in the integral equals 1 on an interval containing $[-2n, 2n]$, so

$$d(\tau_{-\epsilon nu} g_n, O) \geq 2n - R$$

uniformly in u . Therefore

$$\|(\omega_n - \omega_\infty)|_{\mathfrak{A}(O)}\| \leq \int_{-1}^1 h(u) \Psi_\gamma(2n - R) du = \Psi_\gamma(2n - R) \longrightarrow 0.$$

No subnet and no pre-established translation invariance of the limit is used.

5 The conditional two-phase exclusion

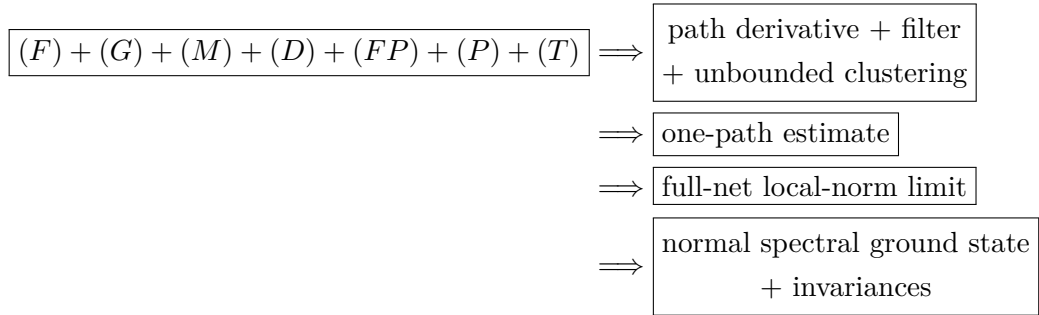
The current Part I theorem correctly makes the following four assumptions explicit:

- (B1) there exist two translation-invariant clustering ground states ω_+, ω_- ;
- (B2) $\theta\omega_+ = \omega_-$;
- (B3) a local order parameter $A = A^*$ has $\omega_\pm(A) = \pm m$, $m \neq 0$;
- (B4) every θ -invariant ground state relevant there is the midpoint $\frac{1}{2}(\omega_+ + \omega_-)$.

The contradiction uses (B4) to identify the locally obtained invariant limit with that midpoint. Without (B4), invariance alone does not imply the identification. GJS 1975–1976 prove phase-transition and pure-phase results in specified strong-coupling models, but the cited pages do not state all four assumptions as a classification of the Hamiltonian ground-state space. Hence the exclusion is rigorous exactly as a conditional theorem and is not an unconditional consequence of Theorem 2.1.

6 Dependency conclusion

The exact dependency chain is



Every arrow has been derived above. The theorem remains conditional on all seven input groups; it is not logically “conditional only” on the two GJS estimates.